

ERBOT Results Summary

CORA Bibliographic Dataset

ERBOT v0.2.0

Generated: 2026-03-26 (rerun) — Run time: $\approx$ 54 minutes

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1 Dataset Overview

Property	Value
Dataset	CORA Bibliographic Bibliography
Records	1,879
Fields	16 (id, title, booktitle, authors, address, date, year, editor, journal, volume, pages)
Task	Deduplication (single-source)
Ground truth	64,578 matched pairs (from cora_gold.csv)

CORA is a standard entity resolution benchmark consisting of research paper citations with substantial noise: missing fields, inconsistent formatting, OCR errors, and partial references. The same real paper may appear multiple times with different field values across entries.

2 Pipeline Configuration

Blocking note: 137,707 pairs from 1,879 records (1,764,381 total possible). The prefix-3 blocking on `title` achieves 92.2% reduction while retaining candidate pairs. Pair completeness (fraction of true matches retained) was not recorded in this run due to truth format — worth measuring in future runs.

Stage	Setting	Value
Blocking	Method	Prefix blocking on <code>title</code> , length 3
Blocking	Candidate pairs	137,707
Blocking	Reduction ratio	0.922 (92.2% of all pairs eliminated)
Similarity	Fields computed	15 (all non-ID fields)
Weights	Method	Equal (1/15 per field)
Clustering	Methods run	9
Merge	Strategy	Best (consensus)

3 Performance Results

All 9 clustering methods evaluated against the gold standard:

Method	ARI	NMI	VI	B ³ -P	B ³ -R	B ³ -F	Hom.	Com.	V	PF-P	PF-R	PF-F
hclust_avg	0.746	0.855	1.54	0.651	0.849	0.737	0.804	0.912	0.855	0.712	0.808	0.757
hclust_ward	0.524	0.827	1.88	0.661	0.748	0.702	0.794	0.863	0.827	0.581	0.508	0.542
pam	0.624	0.841	1.72	0.639	0.784	0.704	0.801	0.885	0.841	0.651	0.626	0.638
svm	0.543	0.849	1.62	0.630	0.817	0.711	0.809	0.893	0.849	0.437	0.803	0.566
threshold_cc	0.506	0.811	1.92	0.520	0.858	0.648	0.733	0.909	0.811	0.394	0.818	0.532
louvain	0.506	0.811	1.92	0.520	0.858	0.648	0.733	0.909	0.811	0.394	0.818	0.532
label_prop	0.506	0.811	1.92	0.520	0.858	0.648	0.733	0.909	0.811	0.394	0.818	0.532
leiden	0.000	0.686	5.18	1.000	0.065	0.122	1.000	0.522	0.686	0.000	0.000	0.000
gc*	0.000	0.000	5.65	0.040	1.000	0.077	0.000	1.000	0.000	0.039	1.000	0.076
<i>final</i>	0.746	0.855	1.54	0.651	0.849	0.737	0.804	0.912	0.855	0.712	0.808	0.757

*Metrics: ARI = Adjusted Rand Index; NMI = Normalized Mutual Information; VI = Variation of Information (lower is better); B³ = B-cubed; V = V-measure; PF = Pairwise F-score; Hom. = Homogeneity; Com. = Completeness. *gc failed with a dimension mismatch error (degenerate fallback).*

4 Final Cluster Assignment

The `final` strategy correctly selected **hclust_avg** — the best-performing method (ARI = 0.746). The merge selection bug present in the original run has been fixed.

- **Total clusters:** 50
- **Records:** 1,879
- **Mean cluster size:** 37.6

50 clusters from 1,879 records is a well-partitioned result, compared to the original buggy run which collapsed all records into just 5 mega-clusters (≈ 900 records each). The current partition is compact and discriminative, reflecting the high-ARI `hclust_avg` solution.

5 Method-by-Method Interpretation

hclust_avg (Best — ARI = 0.746)

The strongest result, and the correct `final` selection. Average-linkage hierarchical clustering on CORA’s 15-field similarity matrix produces a well-balanced partition: B³-precision 0.651, recall 0.849, PairF 0.757. The high recall reflects average-link’s tendency to grow clusters by absorbing nearby records, which works well here because CORA duplicates share many field similarities

and form dense sub-graphs. The 50-cluster solution (mean 37.6 records) is a plausible partition for a dataset of 1,879 citations covering a moderate number of distinct papers.

PAM (ARI = 0.624)

A strong second place. Partitioning Around Medoids at auto-tuned k finds compact, medoid-centred clusters that score well across all metrics (B³-F 0.704, PairF 0.638). PAM is more conservative than `hclust_avg` — slightly lower recall (0.784 vs. 0.849) but higher precision (0.639 vs. 0.651 — near-tie). A good choice when cluster compactness matters.

SVM (ARI = 0.543)

Third place overall. SVM learns a supervised decision boundary and achieves the best VI (1.62) among all methods. Precision (0.630) and recall (0.817) are well-balanced. It is outperformed by `hclust_avg` (+0.203 ARI) and PAM (+0.081 ARI) in this run, suggesting that the 15-field equal-weight similarity score provides a stronger signal for hierarchical and centroidal methods than for the SVM boundary learner at this scale.

Threshold-CC / Louvain / Label Propagation (Identical — ARI = 0.506)

All three converge to exactly the same solution. Good robustness signal but lower ARI than the top three methods. Their higher recall (0.858) trades precision (0.520) against the `hclust/pam` family. At the default threshold of 0.5, graph methods may be over-linking records.

Leiden (Degenerate — ARI = 0.000)

Perfect homogeneity (1.0) but near-zero completeness (0.065) and zero ARI. Leiden over-partitions at the default resolution — splits each true entity into many tiny sub-clusters. Needs a lower resolution parameter (try 0.05–0.2).

GC — Graph Coloring (Failed — ARI = 0.000)

Runtime error: dimension mismatch in the GCIMER interface. Fell back to degenerate output. This is a recurring bug across all datasets that needs fixing.

hclust_ward (ARI = 0.524)

Solid fourth place. Ward linkage is more conservative than average-link — lower recall (0.748) but higher precision (0.661). The ARI gap vs. `hclust_avg` (0.746 vs. 0.524) is notable and suggests average-link better suits CORA’s citation graph structure.

6 Key Findings

1. **Best method is `hclust_avg` (ARI = 0.746).** Average-linkage hierarchical clustering outperforms all other methods on CORA, including SVM (0.543) and PAM (0.624). This is the strongest result across all ERBOT benchmarks to date.
2. **Hierarchical and centroidal methods dominate on CORA.** The top three methods are `hclust_avg` (0.746), PAM (0.624), and SVM (0.543) — all of which build more structured cluster representations than the graph-based threshold-CC family (0.506). CORA’s dense 15-field similarity matrix provides enough information for these methods to find fine-grained partitions.

3. **The final merge is now correct.** After fixing the merge selection bug, `final` correctly identifies `hclust_avg` and reports $\text{ARI} = 0.746$. The original buggy run reported `final` $\text{ARI} = 0.067$ (same method, degenerate 5-cluster output).
4. **Graph-based methods are consistent but not the best here.** Threshold-CC, Louvain, and Label Propagation all converge to $\text{ARI} = 0.506$. Good robustness signal, but the 15-field matrix gives richer signal to hierarchical/centroidal methods.
5. **Equal weights leave room for improvement.** With 15 fields, ARI-based weight learning could further improve `hclust_avg` and SVM — `title` and `authors` are far more discriminative than `note` or `tech`.
6. **Leiden and GC remain broken.** Leiden $\text{ARI} = 0$ (over-partitioning); GC fails with a dimension mismatch error. Neither contributes usable results without fixing.
7. **Blocking is effective.** 92.2% reduction with prefix-3 on `title` is appropriate for bibliographic data where title prefixes are stable across duplicates.

7 Recommendations for Next Steps

Priority	Action
High	Switch from equal weights to supervised ARI-based weights (<code>weights="ari"</code>) — likely to push <code>hclust_avg</code> and SVM above 0.80 ARI.
High	Fix GC dimension mismatch error — it fails silently across all datasets.
Medium	Tune Leiden resolution parameter (try 0.05–0.2) to prevent over-splitting.
Medium	Add pair completeness logging to Stage 3 output (infrastructure is in <code>er_block_stats()</code> , just needs printing).
Low	Try field-specific similarity methods (Jaro-Winkler for <code>authors/title</code> , exact for <code>year</code>) rather than auto-detected uniform similarity.
Low	Try SN blocking (window = 40) instead of prefix-3 — may recover some true matches with corrupted title prefixes.

8 Output Files

File	Description
<code>er_performance.csv</code>	Full metric table (10 methods $\times$ 13 metrics)
<code>er_predictions.csv</code>	Final cluster assignments (1,879 records $\times$ 2 columns)
<code>cora_data_profile.pdf</code>	Dataset profile: field types, distributions, missingness
<code>plot_method_comparison.pdf</code>	Grouped bar chart: ARI, B ³ -F, NMI per method
<code>plot_performance_heatmap.pdf</code>	Colour heatmap: all metrics $\times$ all methods
<code>plot_cluster_sizes.pdf</code>	Cluster size distribution histogram
<code>plot_precision_recall.pdf</code>	Precision vs. Recall scatter (B-cubed and Pair-F)
<code>CORA_RESULTS_SUMMARY.pdf</code>	This document